

Live Argo Overlay

Pick a point, and watch the model's prediction land on top of a real float it has never seen.

PORT	PAGE FILE	MODULE	NETWORK
8508	app/phase2/validate_page.py	phase2/validation/argo_overlay.py	none - fully offline

Every accuracy number in this pack is an average over thousands of comparisons. This page does the opposite: it shows one comparison, in full, live, at a point the audience picks. It is the page built to answer the question a judge is going to ask anyway.

1. What this feature is, in plain words

An **Argo float** is a real instrument drifting in the ocean. Every few days it sinks to a kilometre, then rises while measuring temperature the whole way up, and transmits the profile by satellite. Those measurements are the ground truth in this field.

This page takes a point you choose, runs the **same frozen model** the main dashboard uses - not a second copy, not a re-trained version - and draws its predicted profile with its uncertainty band. Then it drops a real, independent float profile on top and prints the error at every depth.

The float is independent in the strict sense: it comes from the 962 profiles held out of training entirely. Nothing on this page has ever been shown to the model during learning.

The matching is deliberately simple and deliberately explained. Candidates within one degree and five days are ranked by distance first, with time breaking ties. There is no clever weighted score, because there is no defensible exchange rate between kilometres and days, and inventing one would hide an assumption inside a number.

THE EVERYDAY PICTURE

A weather model claiming 31 degrees is one thing. Holding a thermometer up beside it, in front of the room, is another. This page is the thermometer.

2. What you actually see on the screen

One point, one float, one honest comparison:

- The model's predicted profile down 15 depths, drawn with its calibrated plus-or-minus 2 sigma band.
- A real independent Argo float profile drawn on top of it.
- The per-depth error printed live, plus an RMSE and bias summary for that single comparison.
- The float's distance in kilometres and its time offset in days - so the audience can weigh how strong this particular check is.
- A top-k picker, so if several floats are nearby you can step through them rather than being shown only the most flattering one.
- A small map showing where the chosen point landed.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/validate_page.py --server.port 8508
```

- 1 Run the command and open **http://localhost:8508**.
- 2 Enter a latitude and longitude in the basin. Ask a judge to choose it - that is the whole point of this page, and it is safe because nothing here is cached per point.
- 3 Pick a date inside the model's window (1 June 2025 to 23 June 2026).
- 4 Read the float's distance and time offset before reading the error. A float 2 km and same day is a strong check; 60 km and four days is a weaker one, and the page tells you which you have.
- 5 Step through the top-k floats to show you are not selecting the best one.
- 6 Point at the plus-or-minus 2 sigma band and note where the float falls inside it and where it does not - Feature 13 explains why that band is honest rather than flattering.

WHY THE POINT IS TYPED, NOT CLICKED

A click-handler that misfires on demo day is worse than a plain input box that always works. The whole page is offline too - no live float download - so it cannot fail on a conference-hall network.

4. Why the problem statement needs this

The problem statement requires independent validation against float observations. An aggregate RMSE satisfies that requirement on paper; this page satisfies it in the room, where a reviewer can choose the test point themselves.

It also closes the most common objection to any machine-learning result - that the model memorised its training data. A float the model has never seen, at a point the audience picked, is the cleanest available answer to that.

PS requirement	How this feature answers it
Independent validation with Argo	Uses the 962 held-out profiles directly, one at a time, with the match quality on screen.
Report bias and error honestly	Per-depth error is printed live for the single comparison being shown - no aggregation to hide behind.
Reconstruct profile from surface	Runs the shipped inference path itself, so the overlay cannot diverge from the deliverable.

5. The measured numbers behind it

Design properties that make the check defensible [VERIFIED from the module and its tests]:

Property	How it is guaranteed
Same model as the dashboard	The prediction call is a thin wrapper over the shipped predictor. No second model-load path exists , so the overlay cannot silently diverge.
Floats never used in training	Matched against the held-out table of 962 independent profiles - the same set the headline number is scored on.
No hidden interpolation	The float table is pre-binned to the project's 15 standard depths, so model and float already share one depth axis. A depth is scored only where both are finite; the comparison never extrapolates past the float's deepest level.
Deterministic ordering	Ranked by (distance, then absolute time offset, then position and date), so the same query always returns the same float.
Cannot fail on the venue network	Matching is fully offline - 13 tests, all offline, no bundle and no network required.

6. The impact - and who it is for

Who	What changes for them
The jury	The single most convincing live demonstration available - they choose the point, and the model is checked against reality on the spot.
INCOIS / MoES	A routine spot-check tool. Any operator can verify the product against the nearest float before acting on it.
Ocean scientists	Per-depth residuals at a chosen location, which is how model error is actually diagnosed in practice.
The team	A regression alarm: if a checkpoint change breaks something, it shows up here as a visible mismatch rather than a shifted average.

WHY SOCIETY GAINS FROM THIS

Trust in a public forecast product is built by letting people check it. A page where anybody can pick a point and see the prediction measured against a real instrument is the difference between asking the public to believe a system and letting them test it.

7. Say this to the jury

Please choose a point anywhere in the basin. This runs the same frozen model you have already seen, draws its profile with its uncertainty band, and drops a real Argo float on top - one of the 962 profiles the model has never seen. The per-depth error appears live. And notice we show the float's distance and time offset, because a float 2 km away and on the same day is a much stronger check than one 60 km away and four days later. We show you which one you are getting.

Hand this one to the jury. Let them pick the coordinates.

8. Questions the jury will ask

Q. Is this the same model, or a special demo version?

A. The same one. The prediction call is a thin wrapper over the shipped predictor, and there is deliberately no second model-loading path in the codebase - so this page structurally cannot show you a different model from the dashboard.

Q. Could you be picking the float that makes you look best?

A. The ranking is by distance first and time second, fully deterministic, and there is a top-k picker so you can step through every nearby float yourself. You are welcome to pick the worst one.

Q. What if the float falls outside your uncertainty band?

A. It sometimes does, and that is expected and published. Our plus-or-minus 2 sigma band covers 91.2% of held-out observations against a nominal 95.4%, so we are mildly overconfident and we report the coverage as a range by depth rather than as a flattering average. Feature 13 covers that in full.

9. The honest limit - say it before they find it

ONE FLOAT IS ONE FLOAT

A single overlay is a demonstration, not a statistic. The defensible accuracy claim is the aggregate over 962 profiles and 12,736 depth comparisons in Feature 7; this page makes that aggregate tangible but cannot replace it. It is also limited to the held-out window and to points where a float was genuinely nearby - and where none was, it says so rather than showing an empty comparison.

IN ONE LINE

The audience picks the point; the model is measured against a real float it has never seen, with the strength of that check shown on screen.

962

HELD-OUT PROFILES AVAILABLE

2 to 60 km

FLOAT OFFSETS, ALWAYS DISPLAYED

0

SECOND MODEL-LOAD PATHS - IT CANNOT DIVERGE

Feature 8 of 17. Next: Feature 9.